



AI Reasoning Agents

PRESENTED BY
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Problem and Solution



PROBLEM

The increasing demand for **automated reasoning** in AI competitions creates urgency. Current systems struggle with hallucination and formatting errors, leading to unreliable outputs that hinder progress.

SOLUTION

Our proposed **Dual-Agent System** utilizes a Question Agent to generate reasoning problems and an Answer Agent to validate outputs. This ensures high accuracy and structured JSON formatting for reliable results.

System Architecture

Our AI Reasoning Agent utilizes LLaMA 3 8B, fine-tuned with LoRA for optimized performance. Key features include:

- **Data Layer** : Trained on a structured JSON reasoning dataset with controlled input-output formatting.
- **Model Adaptation**: LoRA fine-tuning via the Unsloth framework enables efficient training and retains model capability.
- **Dual-Agent Framework**: The Question Agent creates reasoning problems, while the Answer Agent provides logical solutions.
- **Validation & Control**: A validation layer ensures strict JSON formatting, filters invalid outputs, and controls token generation.

Final Output: Produces clean, validated JSON responses for evaluation and API deployment.



Dataset Preparation



- **Data Collection:** Gathered structured reasoning problems (logical reasoning, puzzles, pattern tasks).
- **Data Cleaning:** Removed duplicates and inconsistencies for dataset integrity.
- **Prompt Structuring:** Created templates for consistent input-output.
- **JSON Schema Design:** Standardized format with fields for question, options, reasoning, and final answer
- **Quality Validation:** Ensured logical correctness and JSON compliance.

Training-Ready Dataset: Optimized dataset prepared for LoRA-based fine-tuning on LLaMA 3 8B.



Inference and Validation



Inference Pipeline

- **Model Loading:** The fine-tuned LLaMA 3 8B model is loaded with LoRA adapters.
- **Tokenization & Prompt Injection:** A well-organized prompt template is utilized before the generation process.
- **Controlled Generation:** Adjusted temperature and decoding parameters ensure consistent reasoning, while stop tokens prevent excessive generation.

Validation Layer

- **Strict JSON Enforcement:** Guarantees that outputs conform to a predefined schema.
- **Malformed Output Filtering:** Removes incomplete or corrupted responses.
- **Hallucination Reduction:** Implements format constraints and structured reasoning prompts.

Output Verification: The final JSON is checked for validity before evaluation or API response.



Results Overview

Enhanced Reasoning Consistency: The logical coherence and accuracy of solutions improved significantly following LoRA fine-tuning.

Structured Output Reliability: Malformed JSON outputs were minimized due to strict validation processes.

Faster Inference Speed: Optimizations with Unsloth led to expedited loading and generation, resulting in quicker responses.

Reduced Hallucination Rate: Controlled decoding effectively minimized unpredictable outputs.

Memory Efficiency: LoRA fine-tuning reduced computational overhead while maintaining the model's capabilities.



Petforence AI-system Key Performance Metrics

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reasoning consistancy



Faster speeds
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